Step-by-Step QuickSort Simulation

Initial Array

We perform QuickSort on the following array using the Lomuto partition scheme (pivot = last element):

$$A = [7, 2, 9, 1, 5, 3, 8, 4, 6, 0]$$

Step 1: quicksort(A, 0, 9), pivot = 0

j	$A[j]$	$A[j] \leq 0$?	Action	Resulting Array
0	7	No	—	—
1	2	No	—	—
2	9	No	—	—
3	1	No	—	—
4	5	No	—	—
5	3	No	—	—
6	8	No	—	—
7	4	No	—	—
8	6	No	—	—

Swap A[0] and A[9] (pivot placement):

$$A = [0, 2, 9, 1, 5, 3, 8, 4, 6, 7]$$

Pivot index = 0. Recurse on:

- quicksort(A, 0, -1)
- quicksort(A, 1, 9)

Step 2: quicksort(A, 1, 9), pivot = 7

j	$A[j]$	$A[j] \leq 7$?	Action	Resulting Array
1	2	Yes	i=1, swap A[1] $\leftrightarrow$ A[1]	—
2	9	No	—	—
3	1	Yes	i=2, swap A[2] $\leftrightarrow$ A[3]	A = [0,2,1,9,...]
4	5	Yes	i=3, swap A[3] $\leftrightarrow$ A[4]	A = [0,2,1,5,9,...]
5	3	Yes	i=4, swap A[4] $\leftrightarrow$ A[5]	A = [0,2,1,5,3,9,...]
6	8	No	—	—
7	4	Yes	i=5, swap A[5] $\leftrightarrow$ A[7]	A = [0,2,1,5,3,4,8,9,...]
8	6	Yes	i=6, swap A[6] $\leftrightarrow$ A[8]	A = [0,2,1,5,3,4,6,9,8,7]

Swap A[7] and pivot (7):

$$A = [0, 2, 1, 5, 3, 4, 6, 7, 8, 9]$$

Pivot index = 7. Recurse on:

- quicksort(A, 1, 6)
- quicksort(A, 8, 9)

Step 3: quicksort(A, 1, 6), pivot = 6

All elements are $\leq$ 6, so just place pivot correctly:

$$A = [0, 2, 1, 5, 3, 4, 6, 7, 8, 9]$$

Pivot index = 6. Recurse on:

- quicksort(A, 1, 5)

Step 4: quicksort(A, 1, 5), pivot = 4

Partitioning on $[2, 1, 5, 3, 4]$, after swaps:

$$A = [0, 2, 1, 3, 4, 5, 6, 7, 8, 9]$$

Pivot index = 4. Recurse on:

- quicksort(A, 1, 3)

Step 5: quicksort(A, 1, 3), pivot = 3

Partitioning on $[2, 1, 3]$, after swaps:

$$A = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]$$

Remaining Calls

All remaining subarrays are size 1 or 0. So the array is fully sorted:

$$\boxed{[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]}$$